

Gerald Mahapranaja Pillian

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Professional Summary

Results-driven Information Systems student and aspiring Data Analytics professional with a strong foundation in computer science, UI/UX design, and data engineering. Demonstrated leadership through international academic exchange at NTUST Taiwan, a published research paper, and multi-disciplinary project involvement across engineering, marketing, and event management.

Education

- National Taiwan University of Science and Technology (NTUST)** August 2025 – January 2026
Exchange Program – Department of Computer Science Taipei, Taiwan
- Developed project: **Fuzzy Logic Control for Dynamic Target Tracking in Two-Axis Autonomous Aiming Systems**
 - Developed project: **Email Communication Simulation System Designs**
 - Developed ML project (Machine Learning Course): **Particle Swarm Optimized Neural Networks for Real-Time Air Quality Forecasting in Taipei's Urban Microenvironments** – trained on **100,000+** samples
- Institut Teknologi Sepuluh Nopember (ITS)** August 2022 – Present
Bachelor of Information Systems Surabaya, Indonesia
- Cumulative GPA: **3.51/4.0** | Runner-up FTEIC 2022 BPC | Semi-Finalist UGMBCC (2023) | Semi-Finalist DANAMON DIBPC (2024)
- Lazuardi High School** October 2019 – October 2022
Natural Sciences (IPA) Track Indonesia
- Graduated with average **89.03**; ranked **7th** in cohort; represented school in international academic competitions

Experience & Involvement

- UI/UX Designer & Developer** May 2024 – June 2024
Data Engineering and Business Intelligence Laboratory, IS ITS Surabaya, Indonesia
- Designed end-to-end UI/UX for **CLASTIC**, an AI-powered plastic waste sorting mobile app; conducted field testing in Sumberbening Village
 - Implemented web and graphic design across digital platforms, ensuring cohesive and user-friendly experiences
- Expert Staff – External Department** April 2023 – August 2025
Information Systems Expo (ISE!), ITS Surabaya, Indonesia
- Progressed from General Staff to Expert Staff; managed and mentored a team of 3 staff members
 - Pitched industry professionals as guests; served as **Master of Ceremonies** for UX Academy and Data Software Academy
 - Achieved a KPI of **300+** participants across organized programs
- Student Staff Volunteer – ITS Global Engagement** December 2023 – February 2024
Institut Teknologi Sepuluh Nopember Surabaya, Indonesia
- Liaised for **100+** international students across **10+** major events; served as speaker, MC, logistics coordinator, and guide
- Assistant Lecturer – Introduction to Information Systems Concept** September 2023 – December 2023
Department of Information Systems, ITS Surabaya, Indonesia
- Designed assignments and projects; provided grading, academic support, and coordination with course lecturers
- Strategy & Development Staff** June 2023 – February 2024
IEEE, ITS Surabaya Student Branch Surabaya, Indonesia
- Principal editor for marketing and archival content; completed **5+** major projects on the Documentation and Editorial Team

Research & Publications

- Published Paper | ResearchGate** 2024
- “*Understanding User Perceptions of Camera and Microphone Access in Mobile Applications: Exploring Permission Granting and Associated Security and Privacy Risks*”
 - Investigated user trust patterns and security/privacy implications of app permission systems on mobile platforms

Notable Projects

- Particle Swarm Optimized Neural Networks for Air Quality Forecasting** NTUST, 2025
- Built and trained a neural network on **100,000+** samples using PSO for hyperparameter tuning to predict real-time urban air quality
- Fuzzy Logic Control for Dynamic Target Tracking in Two-Axis Autonomous Aiming Systems** NTUST, 2025
- Designed a fuzzy logic controller for real-time dual-axis target tracking; applied control theory and computational logic
- Email Communication Simulation System Designs** NTUST, 2025
- Engineered a simulation of email communication protocols; focused on system architecture and server-client flow modelling
- CLASTIC – AI-Based Plastic Waste Sorting Mobile Application** ITS, 2024
- Led UI/UX design for an AI-powered waste management app deployed in rural Indonesia; integrated ML-based plastic classification

Certifications

SAP Certified Application Associate <i>SAP BusinessObjects BI Platform 4.2 SAP University Alliances</i> 3-month ERP integration program covering SD, MM, and PP modules; signed by Mr. Crispian Tan, SAP University Alliances	Issued July 2024 <i>Online</i>
Data Analyst Certification <i>MySkill</i> Hands-on training in SQL, Python, and DBMS; built and presented a guided capstone data analysis project	August – October 2023 <i>Online</i>

Technical Skills

Programming: Python, SQL, JavaScript, HTML/PHP (Laravel), Java, VBA
Data & ML: R Studio, DBMS/SQL, Machine Learning (Neural Networks, PSO), SAP ERP & BI Platform
Web & App Dev: Supabase (Database, Auth, APIs), Google Cloud APIs, REST API Integration, PHP/Laravel
Tools: Microsoft Office (Advanced), Google Workspace (Advanced), GDevelop, MicroStudio
Design & Media: UI/UX Design, Video Editing (Advanced), Professional Photography (Advanced), Photo Editing, General Marketing
Soft Skills: Leadership, Public Speaking, Event Management, Problem Solving, Adaptability, Emotional Intelligence

Languages

Indonesian – Native	Javanese – Intermediate	Arabic – Basic	German – Basic
English – Fluent	Mandarin – Basic	French – Basic	